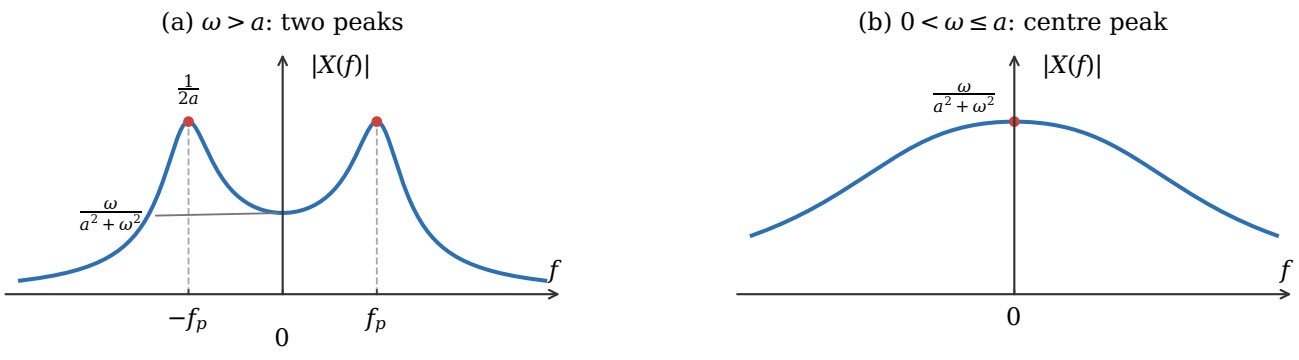

$f_p = \sqrt{\omega^2 - a^2}/(2\pi)$; shape schematic, not to scale.